Infinite series

Math 163 • Prof. Chowdhury • Concept map

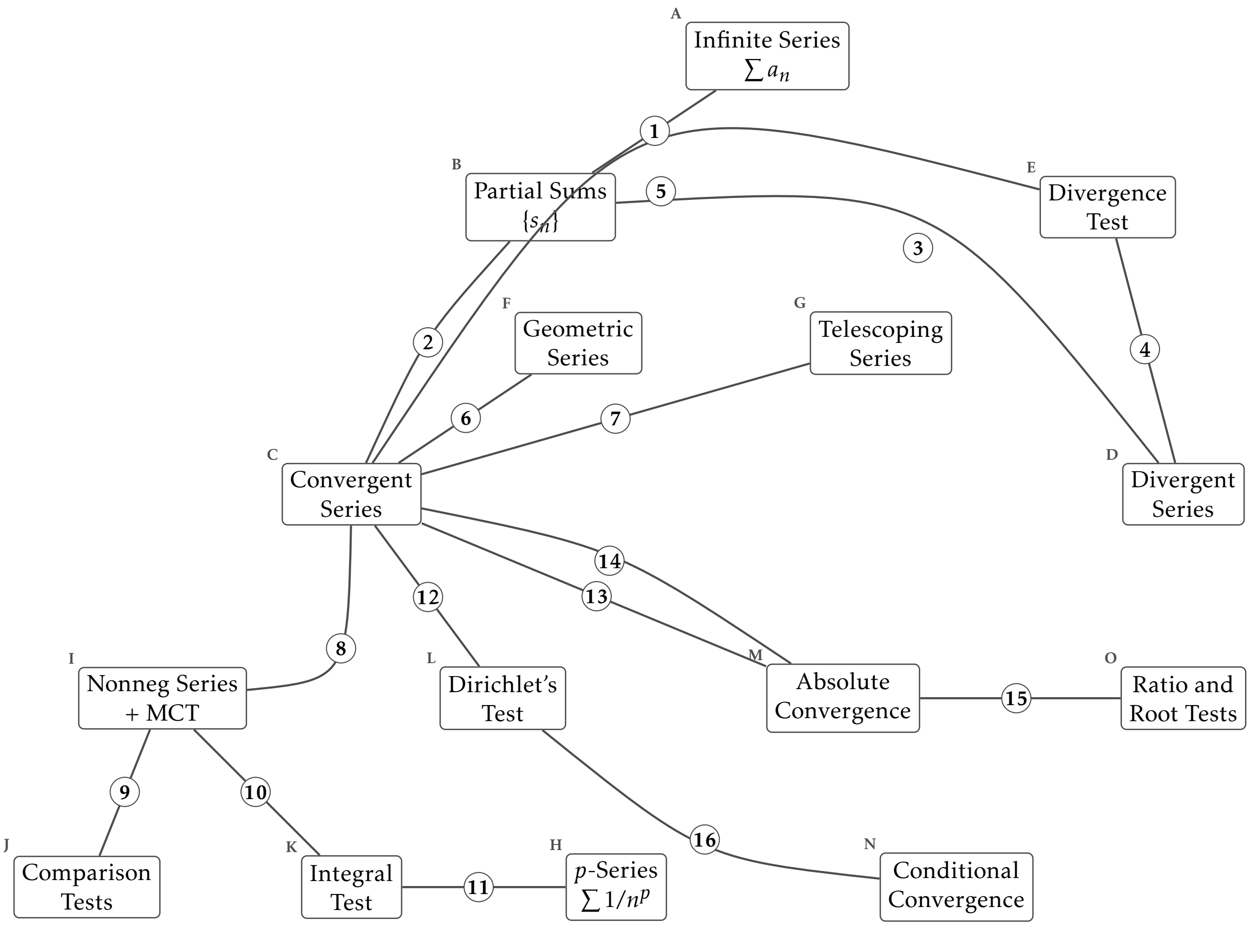
The next page shows 15 ideas as labelled boxes, joined by 16 numbered arrows. Every arrow is a real mathematical relationship, and the drawing does not tell you which kind: an arrow may *hold*, it may *fail*, the two ideas may be *equivalent*, or it may hold on *other hypotheses* than the ones its neighbours use. For each numbered arrow, circle the kind and describe the relationship in one sentence. Name the theorem, state the hypothesis that is doing the work, or say what goes wrong.

What each box means

- **A. Infinite Series $\sum a_n$.** The expression $a_1 + a_2 + a_3 + \cdots$. It has no meaning until the partial sums give it one.
- **B. Partial Sums $\{s_n\}$.** $s_n = a_1 + \cdots + a_n$. An ordinary sequence, so everything you know about sequences applies to it.
- **C. Convergent Series.** $\sum a_n$ converges when $\{s_n\}$ has a finite limit, and the sum is that limit.
- **D. Divergent Series.** $\sum a_n$ diverges when $\{s_n\}$ fails to converge, whether it runs away or oscillates.
- **E. Divergence Test.** If $a_n \not\to 0$ then $\sum a_n$ diverges. It is a test for divergence only, and it never certifies convergence.
- **F. Geometric Series.** $\sum ar^n$, which converges exactly when $|r| < 1$, with sum $a/(1-r)$ and partial sums $a(1-r^n)/(1-r)$.
- **G. Telescoping Series.** $\sum(b_n - b_{n+1})$, whose partial sums collapse to $b_1 - b_{n+1}$.
- **H. p-Series $\sum 1/n^p$.** Converges exactly when $p > 1$. The boundary $p = 1$ is the harmonic series, which diverges.
- **I. Nonneg Series + MCT.** If $a_n \geq 0$ the partial sums increase, so the monotone convergence theorem turns convergence into boundedness.
- **J. Comparison Tests.** Direct: $0 \leq a_n \leq cb_n$ eventually and $\sum b_n$ convergent gives $\sum a_n$ convergent. Limit: $\lim a_n/b_n \in (0, \infty)$ gives the same behaviour for both.
- **K. Integral Test.** If f is positive, continuous and eventually decreasing with $f(n) = a_n$, then $\sum a_n$ and $\int_1^\infty f$ converge or diverge together.
- **L. Dirichlet's Test.** If a_n decreases to 0 and the partial sums of b_n are bounded, then $\sum a_n b_n$ converges. Proved by summation by parts.
- **M. Absolute Convergence.** $\sum a_n$ converges absolutely when $\sum |a_n|$ converges. This is strictly stronger than convergence.
- **N. Conditional Convergence.** $\sum a_n$ converges conditionally when it converges and $\sum |a_n|$ does not. The convergence rests on cancellation.
- **O. Ratio and Root Tests.** With $L = \lim |a_{n+1}/a_n|$ or $L = \limsup |a_n|^{1/n}$: absolutely convergent if $L1$, and no conclusion if $L = 1$.

Examples worth keeping to hand

- The harmonic series $\sum 1/n$ diverges even though its terms tend to zero.
- The alternating harmonic series $\sum(-1)^{n+1}/n$ converges conditionally, to $\ln 2$.
- $\sum \sin(n\theta)/n$ converges by Dirichlet's test and is not alternating.
- Summation by parts is the discrete integration by parts, and it is what proves Dirichlet's test.
- The Cauchy criterion for series: control every far enough tail at once, not one chosen tail.



Every arrow is drawn the same way. For each one, decide whether it holds, fails, is an equivalence, or holds on other hypotheses, and say what it claims.

The arrows

1. holds / fails / equivalent / other hypotheses
   ....

2. holds / fails / equivalent / other hypotheses
   ....

3. holds / fails / equivalent / other hypotheses
   ....

4. holds / fails / equivalent / other hypotheses
   ....

5. holds / fails / equivalent / other hypotheses
   ....

6. holds / fails / equivalent / other hypotheses
   ....

7. holds / fails / equivalent / other hypotheses
   ....

8. holds / fails / equivalent / other hypotheses
   ....

9. holds / fails / equivalent / other hypotheses
   ....

10. holds / fails / equivalent / other hypotheses
    ....

11. holds / fails / equivalent / other hypotheses
    ....

12. holds / fails / equivalent / other hypotheses
    ....

13. holds / fails / equivalent / other hypotheses
    ....

14. holds / fails / equivalent / other hypotheses
    ....

15. holds / fails / equivalent / other hypotheses
    ....

16. holds / fails / equivalent / other hypotheses
    ....

One last question

Which single arrow carries the most of this chapter? Say why in one sentence. There is more than one defensible answer, and the argument is the useful part.

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